# STAA57 Week-5

Shahriar Shams

## Learning goals

- Descriptive statistics
- Idea of statistical distribution
- Some common continuous distributions
- Some common discrete distributions
- Idea of population, sample, parameter, statistic
- Estimate vs estimator
- Sample mean as an estimator of population mean

# Descriptive statistics

- In real life, if money and time (resource in general) is not an issue, we can collect data about any variable that we are interested in. For example,
  - Income of all Canadians
  - Number of trees in Ontario
  - Amount of rainfall in Ontario in a certain period of time
- Once we have collected that information, we will probably want to summarize them as it is not possible to remember or use millions of rows of data.
- Few summaries that we are generally interested are
  - Mean/ Average
  - Median
  - Mode
  - Variance (or standard deviation)
  - Quantiles

## Mean/Average

- When we have finite amount of data points(say $X_1, X_2, ..., X_N$) we define mean as

$$mean = \frac{1}{N}\sum_{i=1}^{N} X_i$$

- Suppose we have the following vector data. And we want to calculate the mean

```
x=c(1,1,1,1,2,2,2,3,3)
mean(x)
```

```
## [1] 1.777778
```

- Mean/Average is also called a measure of the center of the data.
- When we have a large *outlier* (a value too high or too low compared to the rest of the data points), mean/average changes significantly making it not representative of the rest of the points.

```
x=c(1,1,1,1,2,2,2,2,3,300)
mean(x)
```

```
## [1] 31.5
```

- That's why some time we calculate *trimmed* mean which is calculating the mean by discarding a certain percentage of data from the two ends.

```
x=c(1,1,1,1,2,2,2,2,3,300)
mean(x,trim = 0.1)
```

```
## [1] 1.75
```

- Be careful when calculating mean in R as if there is a missing value R does not calcualte the mean automatically.

```
x=c(1,1,1,1,2,2,2,2,3,NA)

#this following line will return NA
mean(x)
```

```
## [1] NA
```

```
#this following will give the mean removing the NA's
mean(x,na.rm=TRUE)
```

```
## [1] 1.666667
```

## Median

- Median is another measure of the center which is often used and preferred over mean if we have some potential outliers in the data.
- By definition, Median is the value which divides our data points into two halves. 50% of the observations will be below the median and 50% of the observations will be above the median.

```
x=c(1,1,1,1,2,2,2,2,3,300)
median(x)
```

```
## [1] 2
```

```
x=c(1,1,1,1,2,2,2,2,3,NA)
median(x,na.rm = TRUE)
```

```
## [1] 2
```

## Mode

- Mode is the data point that occurs the maximum number of time.
- For example, in the list of (1,1,1,1,2,2,2,3,3), the mode is 1.

## Variance

- Along with the center of the data, often we are interested in the *spread* of the data.
- A commonly used measurement of the spread of the data is standard deviation which measures on an average how far our data points are from the center of the data.
- Suppose $\mu$ is the mean(center) of the data

$$variance = \frac{1}{N}\sum_{i=1}^{N}(X_i - \mu)^2$$

- The formula above is for population level data. In case you have sample and you want to calculate the variance from your sample the formula is slightly different.
- Standard deviation is the square root of the variance.

```
x=c(1,1,1,1,2,2,2,3,3)

mu=mean(x)

pop.variance = mean((x-mu)^2)

pop.sd = sqrt(pop.variance)


# these following two lines will calculate the variance and
# standard deviation considering it is a sample
var(x)
```

```
## [1] 0.6944444
```

```
sd(x)
```

```
## [1] 0.8333333
```

## Quantiles

- Quantiles can be seen as an extension of the idea of median.
- Median divides the data in half and half.
- The qth quantile (or q*100 percentile) gives a value which divides our data in q*100% on the left and (1-q)*100% on the right.

```
x=c(1,1,1,1,2,2,2,3,3)

quantile(x,probs=c(0.25,0.5,0.75))
```

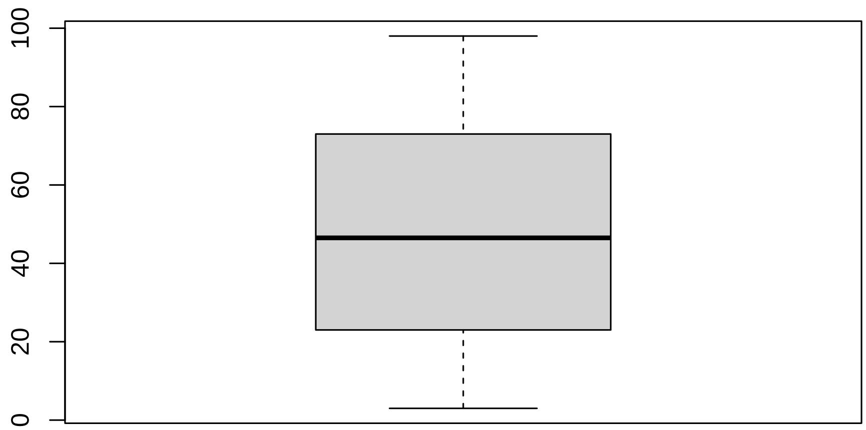
```
## 25% 50% 75%
##   1   2   2
```

## Box plot

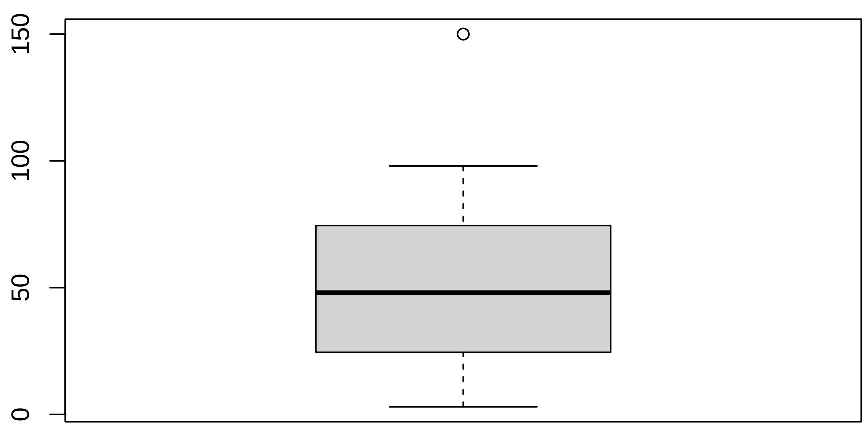
- A simple plot the summarizes the shape of the data.
- The box represents the 0.25quantile, median and 0.75 quantile.
- The two ends are calculated as function of the quantiles. Leaving these calculations for a future course.

```
x=sample(c(1:100),size = 30)
```

```
boxplot(x)
```




```
boxplot(c(x,150))
```



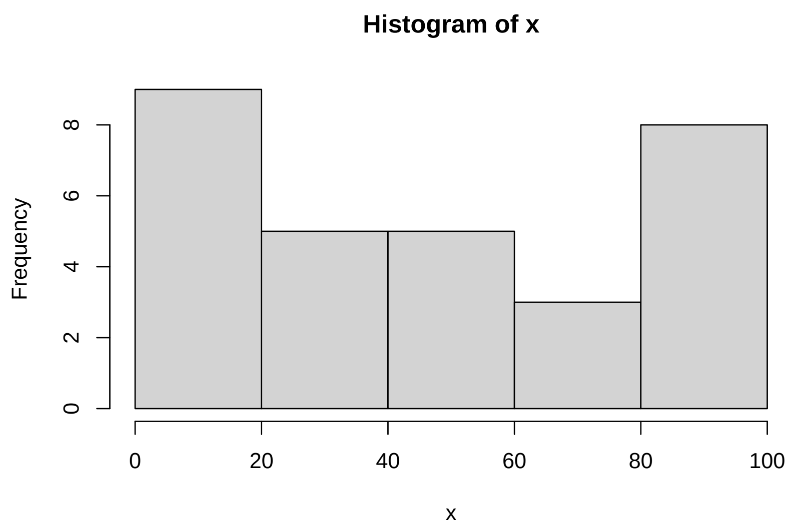

 

## Histograms

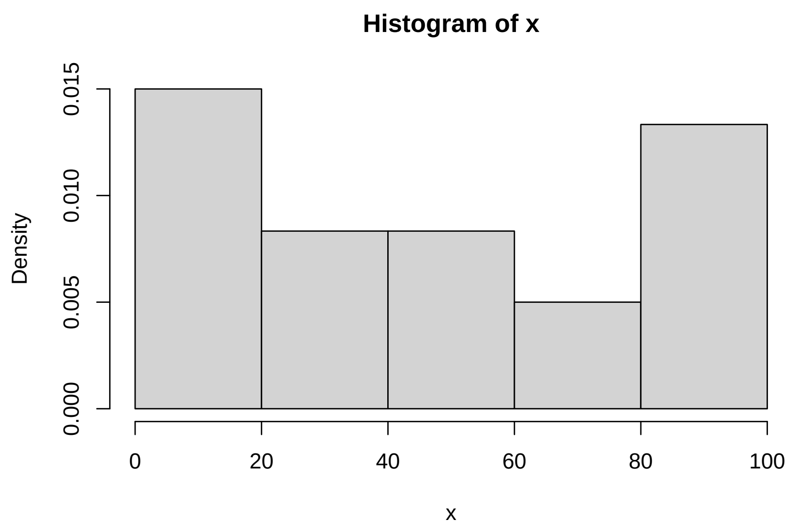
- Histograms are one of the most widely used graphical representation of data.
- Once we have sorted data, we divide the range of the data (min to max) in predefined intervals (we call them bins).
- We count how many observations fall inside each bin.
- we produce a graph where a bar is placed on top of each bin and the height of a graph represents the count of observations falling into each bin.

```
x=sample(c(1:100),size = 30)

#regular histogram
hist(x)
```




```
#density histogram
hist(x,freq=FALSE)
```



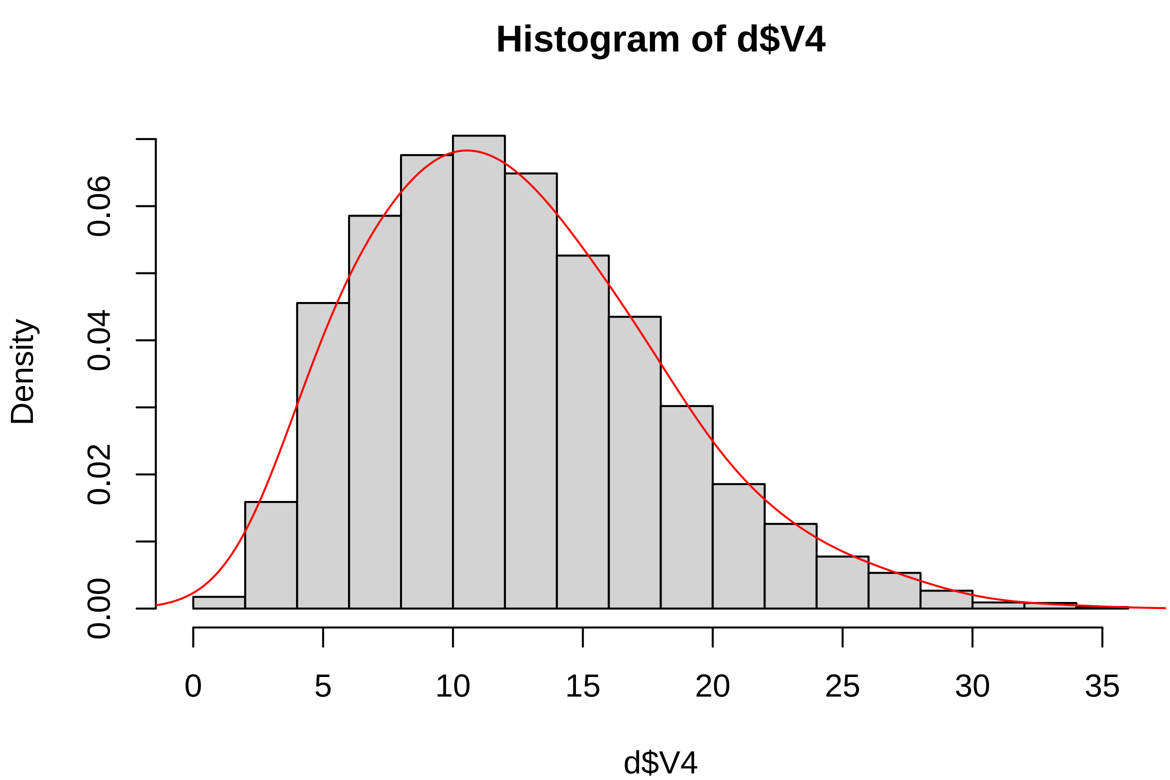
# Example of probability distribution using real data

- We will use a real life data available at http://lib.stat.cmu.edu/datasets/
- In the data set we have daily average wind speeds for 1961-1978 at 12 synoptic meteorological stations in the Republic of Ireland.

```
d=read.table(file="http://lib.stat.cmu.edu/datasets/wind.data",header=F)
head(d)
```

```
##   V1 V2 V3    V4    V5    V6    V7    V8   V9   V10   V11   V12   V13   V14
## 1 61  1  1 15.04 14.96 13.17  9.29 13.96 9.87 13.67 10.25 10.83 12.58 18.50
## 2 61  1  2 14.71 16.88 10.83  6.50 12.62 7.67 11.50 10.04  9.79  9.67 17.54
## 3 61  1  3 18.50 16.88 12.33 10.13 11.17 6.17 11.25  8.04  8.50  7.67 12.75
## 4 61  1  4 10.58  6.63 11.75  4.58  4.54 2.88  8.63  1.79  5.83  5.88  5.46
## 5 61  1  5 13.33 13.25 11.42  6.17 10.71 8.21 11.92  6.54 10.92 10.34 12.92
## 6 61  1  6 13.21  8.12  9.96  6.67  5.37 4.50 10.67  4.42  7.17  7.50  8.12
##     V15
## 1 15.04
## 2 13.83
## 3 12.71
## 4 10.88
## 5 11.83
## 6 13.17
```

```
hist(d$V4,freq=FALSE)
lines(density(d$V4,adjust=2),col="red")
```



- Things that happen naturally usually have a pattern as we have seen in this demonstration.
- We can try to find the functional form of the curve that fits the histogram and that will be statistical distribution that can be applied to wind speed data (at least for Ireland).

# Some common continuous distributions

## Definition of continuous random variable

A *random variable* is *continuous* if it takes *infinite* number of values that are **not countable**.

Example:

- Height of a person.
  - It can be 170.66cm or can be 170.6666666.... cm
  - we have infinite numbers (uncountable) between 170cm and 171cm.
- Waiting time for the next bus.
  - It can be 5.5 minutes or 5.534678023... minutes
  - we have infinite numbers (uncountable) between 5.5 min and 5.6 min.
- Other examples: price of something, age, weight, area, volume, health care cost, income, expense etc.

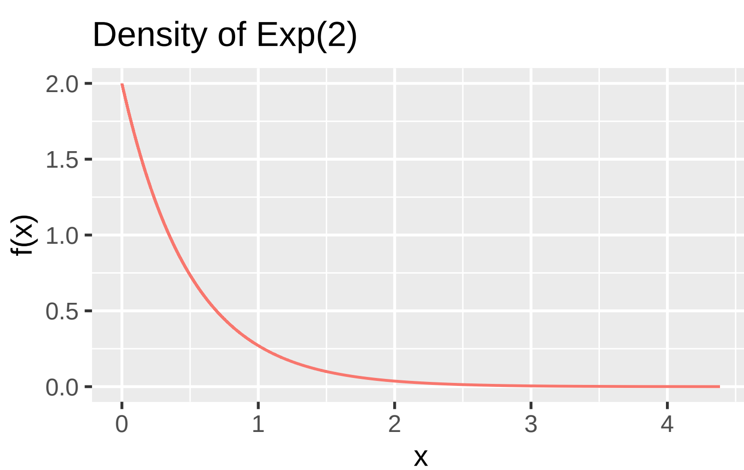
## Exponential distribution

A continuous random variable $X$ has an *Exponential* distribution with parameter $\lambda$ if its *pdf* is defined as

$$f(x) = \lambda e^{-\lambda x} \quad for \;\; x \geq 0$$

In notation, we say $X \sim Exp(\lambda)$

**Note:** $\lambda$ is the "rate parameter" and $\lambda > 0$




- Suppose $\lambda = 2$, so $f(x) = 2e^{-2x}$.

- For continuous random variable, probability is defined as the area under the $f(x)$.
- So for our example, $P[X < 1] = \int_0^1 2e^{-2x}dx$

- Median of a distribution can be defined as "m" that satisfies $\int_{-\infty}^m f(x)dx = 0.5$
- Exercise: can you calculate the median for Exponential distribution with $\lambda = 2$?

- Lets calculate these things using R

```
#P[X<1]
pexp(1,rate=2)
```

```
## [1] 0.8646647
```

```
#median
qexp(0.5,rate=2)
```

```
## [1] 0.3465736
```

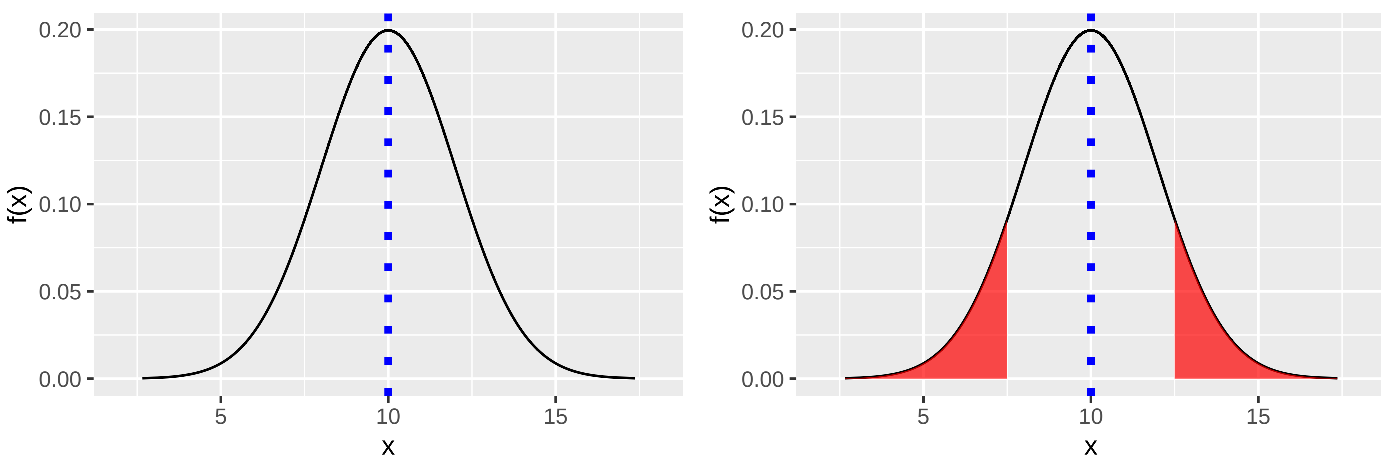
## Normal distribution

A continuous random variable $X$ has a *Normal* distribution with parameter $\mu$ and $\sigma^2$ if its *pdf* is defined as

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad for \quad -\infty < x < \infty\,;\ \sigma^2 > 0$$

In notation, we say $X \sim N(\mu, \sigma^2)$

#### Density of a $N(\mu = 10, \sigma^2 = 4)$




#### Properties of Normal distribution:

- $\mu$ is the center of density (the dotted line on the graphs)
- $\sigma^2$ represents the spread of the density
- $\mu$ and $\sigma^2$ represent mean and variance of this distribution which we will learn next week.
- Normal density is symmetric around $\mu$ (the two shaded regions have the same probability)

## Probability calcualtions using R

```r
# X~N(mu=10, sd=2). Calculate P[X<5]
pnorm(5, mean=10,sd=2)
```

```
## [1] 0.006209665
```

```r
# X~N(mu=10, sd=2). Calculate P[X<12]
pnorm(12, mean=10,sd=2)
```

```
## [1] 0.8413447
```

```r
# X~N(mu=10, sd=2). Calculate P[5<X<12]
pnorm(12, mean=10,sd=2) - pnorm(5, mean=10,sd=2)
```

```
## [1] 0.8351351
```

## Calculating quantiles using R

```r
# 0.05 quantile or 5th percentile
qnorm(0.05,mean=10,sd=2)
```

```
## [1] 6.710293
```

```r
# median or 0.5 quantile or 50th percentile
qnorm(0.5,mean=10,sd=2)
```

```
## [1] 10
```

```r
# 0.95 quantile or 95th percentile
qnorm(0.95,mean=10,sd=2)
```

```
## [1] 13.28971
```

## Standard Normal Distribution

A Normal distribution with $\mu = 0$ and $\sigma^2 = 1$ is called the standard normal distribution.

Conventionally, a random variable following a standard normal distribution is denoted by $Z$

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \quad for \quad -\infty < z < \infty$$

In notation, we write $Z \sim N(0,1)$

## Relationship between Normal and standard Normal

Let $X \sim N(\mu, \sigma^2)$.

Let $Z$ be a transformation $Z = \frac{X-\mu}{\sigma}$

Then, $Z \sim N(0,1)$

# Some common discrete distributions

## Discrete random variable

- A random variable that takes *countable* number of values is called a discrete random variable. Here are few examples:
  - Faces of a die
  - Number of success out of n trial
  - Number of accidents on 401 in any week
  - Number of trees in Ontario

## Bernoulli Distribution:

A random variable $X$ has a *Bernoulli* distribution with parameter $\theta$ (it's a number between 0 and 1) if its probability function is

$$P[X = a] = \begin{cases} \theta & when \quad a = 1 \\ 1 - \theta & when \quad a = 0 \end{cases}$$

In notation, we say $X \sim Bern(\theta)$

## Binomial Distribution:

- Binomial is the sum of $n$ independent Bernoulli random variables.
- Suppose $X_1, X_2, ..., X_n$ are all independent of each other.
- Individually they all have a $Bern(\theta)$ distribution.
- Let $Y$ be another random variable where, $Y = \sum_{i=1}^{n} X_i$
- $Y$ will have a Binomial distribution with parameter $n$ and $\theta$
- In notation, $Y \sim Bin(n, \theta)$

then probability function of $Y$ is given by

$$P[Y = k] = \left( \begin{array}{c} n \\ k \end{array} \right) \theta^k (1 - \theta)^{n-k} \text{ where, } k = 0, 1, 2, ..., n$$

Here,

$$\left( \begin{array}{c} n \\ k \end{array} \right) = \frac{n!}{k!(n-k)!}$$

**Example:**

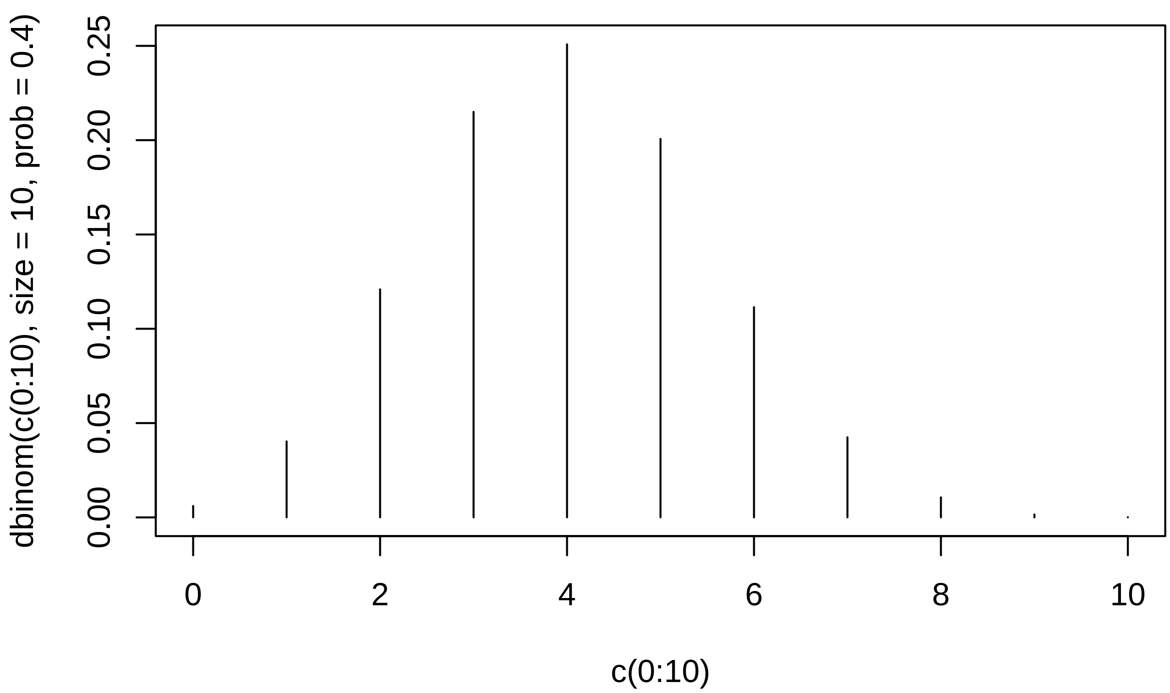
- Tossing a coin $n = 3$ times (or tossing three independent coins)
- Let $X_1$ be the outcome of the first toss with $X_1 = 1$ if it's a head , 0 otherwise
- Similarly $X_2$ and $X_3$ represent the second and the third toss.
- Each time the probability of head is $p$
- we can write $X_1 \sim Bern(\theta)$, $X_2 \sim Bern(\theta)$ and $X_3 \sim Bern(\theta)$
- Let Y = the number of heads in $n = 3$ tosses $\implies Y = X_1 + X_2 + X_3$

$$
\begin{aligned}
P(Y=1) &= P(X_1+X_2+X_3=1) \\
&= P\left(Y=1+0+0 \; or \; Y=0+1+0 \; or \; Y=0+0+1\right) \\
&= \theta(1-\theta)^2+\theta(1-\theta)^2+\theta(1-\theta)^2 \\
&= 3\theta(1-\theta)^2 \\
&= \left(\begin{array}{c} 3 \\ 1 \end{array}\right)\theta^1(1-\theta)^{3-1}
\end{aligned}
$$

$\left(\begin{array}{c} n \\ k \end{array}\right)$ gives us the number of ways we can have a sum of $k$ out of $n$ Bernoulli variables.

**Calculating probability under Binomial distribution using R**

```r
# Binomial(10,0.4) probability function plot
plot(c(0:10),dbinom(c(0:10),size=10,prob=0.4),type="h")
```




```r
# For Bin(10, 0.4); calculating P[X=4]
dbinom(4, size=10, prob = 0.4)
```

```
## [1] 0.2508227
```

```r
#
#
#
#
```

```r
# For Bin(10, 0.4); calculate P[X = 4 or 5]
# Any of these following lines will work.
dbinom(4, size=10, prob = 0.4) + dbinom(5, size=10, prob = 0.4)
```

```
## [1] 0.4514808
```

```r
sum(dbinom(c(4,5), size=10, prob = 0.4))
```

```
## [1] 0.4514808
```

```r
pbinom(5, size=10,prob=0.4) - pbinom(3, size=10,prob=0.4)
```

```
## [1] 0.4514808
```

# Idea of population, sample, parameter, statistic

- We will define all these terms using a simple example.
- Suppose we want to know the **average height** of **ALL the students of UofT** (let's call it $\mu$).
- Due to resource constraint, a sensible approach will be to take a sample of 100 students (or some other number) and measure their average height (let's call it $\bar{X}$ which represents the sample mean).
- "Population" is defined as the collection of all subjects/individuals that we want to make a comment on.
  - in our example, **Population:** All the students of UofT (some 60,000 of us)
- "Parameter" is defined as any summary of the population. It's an unknown number believed to be a constant.
  - in our example, **Parameter:** The average height of the population, which we represent using $\mu$.
  - Instead of population mean, we can be interested in population variance, median etc. Each of these summaries can be treated as a parameter.
- "Sample" is defined as any subset of the population that we actually get to observe.
  - in our example, **Sample:** 100 randomly selected students.
- "Statistic" is defined as any summary of the sample.
  - in our example, **Statistic:** the sample mean.
  - Instead of calculating the sample mean, we can calculate many other summaries such as sample variance, minimum value of the sample, maximum value of the sample. Each of these are called a statistic.

# Estimator vs Estimate

- When a "statistic" is used to make an educated guess about the unknown "parameter", we call this "statistic" either an estimate or an estimator based on whether we have already observed the sample observations or not.
- Say sample mean is the statistic that we are using to guess the value of $\mu$.
- If we have already observed our sample of size $n$, then we have $(x_1, x_2, ..., x_n)$
- We will be able to get a numeric value of the sample mean(statistic).
  - this numeric value is called an estimate. we will write it as $\bar{x}$
  - $\bar{x}$ is an **estimate** of $\mu$
- If we haven't observed our samples yet, the observations are random variables and hence written as $(X_1, X_2, ...X_n)$.
- In theory the sample mean can vary from one set of sample of size $n$ to another set of sample of size $n$.

- Hence, the sample mean is considered as a random variable and written as $\bar{X}$
  - $\bar{X}$ is an **estimator** of $\mu$
- In summary, *Estimate* ($\bar{x}$) is a fixed numeric number, *Estimator* ($\bar{X}$) is a random variable.

# Sample mean as an estimator of population mean

## variation within sample mean

- R demonstration